

Shantanu Shastri

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EDUCATION

Carnegie Mellon University	December 2026
Master of Information Systems Management	GPA: 3.85
Manipal Institute of Technology	August 2022
Bachelor of Electronics and Communications Engineering	GPA: 3.99
Relevant Coursework: Data Structures, Algorithms, Distributed Systems, OOP, System Design	

SKILLS

Languages: TypeScript, Python, Java, C++, JavaScript, SQL, Bash/Shell
Frontend & UI: React, Component Systems, Angular, Visual & Interaction Design, Clean Frontends
Backend & APIs: FastAPI, API Design, Internal Tooling, System Architecture, Node.js, Spring Boot
Data Modeling & Schema: PostgreSQL, Database Design, Data Pipelines, Redis, Neo4j, Milvus, Sybase
Infrastructure: Docker, Apache Kafka, Zookeeper, Kubernetes, Azure, AWS, CI/CD, Observability

EXPERIENCE

BlackRock	January 2022 – August 2026
AI Engineer Internship	June 2026 – August 2026
Data Pipelines & Integrations: Built a fault-tolerant Python FastAPI backend and robust data pipelines (Kafka, PostgreSQL) to connect the AI platform with external trade systems for stream-based ingestion. Schema Design & Infrastructure: Designed clean, well-typed database schemas and infrastructure, linking Milvus embeddings to Neo4j nodes via a PostgreSQL bridge to resolve data inconsistencies. Shared Libraries & APIs: Engineered internal tooling and shared component systems backed by strictly typed RESTful APIs, enabling the engineering team to safely deploy complex AI capabilities. Platform Observability: Built extensive observability workflows to track agent latency and data drift, taking end-to-end ownership to process 3.5M records and handle \$14T in cash.	
Software Developer 2 - Associate	January 2025 – July 2025
TypeScript User-Facing Experiences: Engineered well-modeled TypeScript micro-frontends to translate natural language into JQL, focusing on clean component design and maintainable frontend architectures. System Architecture: Architected scalable Spring Boot data framework and TTL Azure microservices from scratch, managing complex dependencies and reducing infrastructure storage costs by 40%.	
Software Developer 1 - Analyst	July 2022 – December 2024
Full-Stack Interface Design: Spearheaded end-to-end development of responsive UIs integrating Azure's LLMs directly into user workflows, applying strong visual judgment and clean interaction design. Database Optimization & Modeling: Took end-to-end ownership of database optimization, refactoring PostgreSQL schemas on tables exceeding 50M+ rows to achieve a 37% reduction in query latency. Rigorous Engineering Standards: Built comprehensive Azure DevOps CI/CD pipelines with integrated quality gates and JUnit testing suites, ensuring robust, well-documented full-stack deployments.	

PROJECTS

High-Performance Modular Booking Architecture
Designed clean database schemas and decoupled microservices using Apache Kafka, seamlessly handling bursts of 100,000+ real-time requests with minimal latency spikes and containerized deployments.
Zulip Open-Source GenAI Extensions
Built a high-throughput message engine using robust data modeling in Node.js and Redis to process live data streams into concise summaries, prioritizing clean API contracts and zero message loss.